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HEART FAILURE AND CARDIOMYOPATHIES

CLINICAL CASE

Artificial Intelligence-Based Electrocardiographic Analysis Facilitated Diagnosis of Hypertrophic Cardiomyopathy With Successful Treatment Using Mavacamten

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ABSTRACT

BACKGROUND Despite a prevalence of 1 in 500, it is estimated that 85% patients with hypertrophic cardiomyopathy (HCM) may be underdiagnosed or misdiagnosed. There are emerging data on role of artificial intelligence enhanced 12-lead electrocardiogram (AI-ECG) in detection of HCM.

CASE SUMMARY In the current report, we describe the case of a 61-year-old man who was diagnosed with obstructive HCM using AI-ECG, performed as part of an executive physical with subsequent successful initiation and therapeutic response to mavacamten.

DISCUSSION An earlier and accurate diagnosis of HCM is a major unmet need and there are major efforts to enhance the ability a clinician to diagnose HCM as early as possible to be able to institute optimal therapies.

TAKE-HOME MESSAGES The case describes the use of AI-ECG to diagnose HCM appropriately in a 61-year-old symptomatic patient with no previous knowledge of the disease. Following an appropriate diagnosis of HCM, the patient was initiated on the novel cardiac myosin inhibitor mavacamten, with an excellent clinical response. (JACC Case Rep. 2025;30:103228) © 2025 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Hypertrophic cardiomyopathy (HCM) is a commonly inherited cardiomyopathy with a prevalence of 1 in 500, but a varied phenotypic expression ranging from asymptomatic to heart failure to sudden cardiac death (SCD), which occurs in <1% per year.[1,2] A characteristic finding in HCM is dynamic left ventricular outflow tract (LVOT) obstruction (observed in ~70% of patients with HCM, if looked for carefully) and concomitant mitral regurgitation caused by systolic anterior

TAKE-HOME MESSAGES

- The case describes the use of AI-ECG to appropriately diagnose HCM in a 61-year-old symptomatic patient with no previous knowledge of the disease.
- Following an appropriate diagnosis of HCM, patient was initiated on the novel cardiac myosin inhibitor mavacamten, with an excellent clinical response.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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ABBREVIATIONS AND ACRONYMS

AI-ECG = artificial intelligence enhanced 12-lead electrocardiogram

CAC = coronary artery calcium

HCM = hypertrophic cardiomyopathy

LVEF = left ventricular ejection fraction

LVOT = left ventricular outflow tract

METs = metabolic equivalents

SRT = septal reduction therapy

SCD = sudden cardiac death

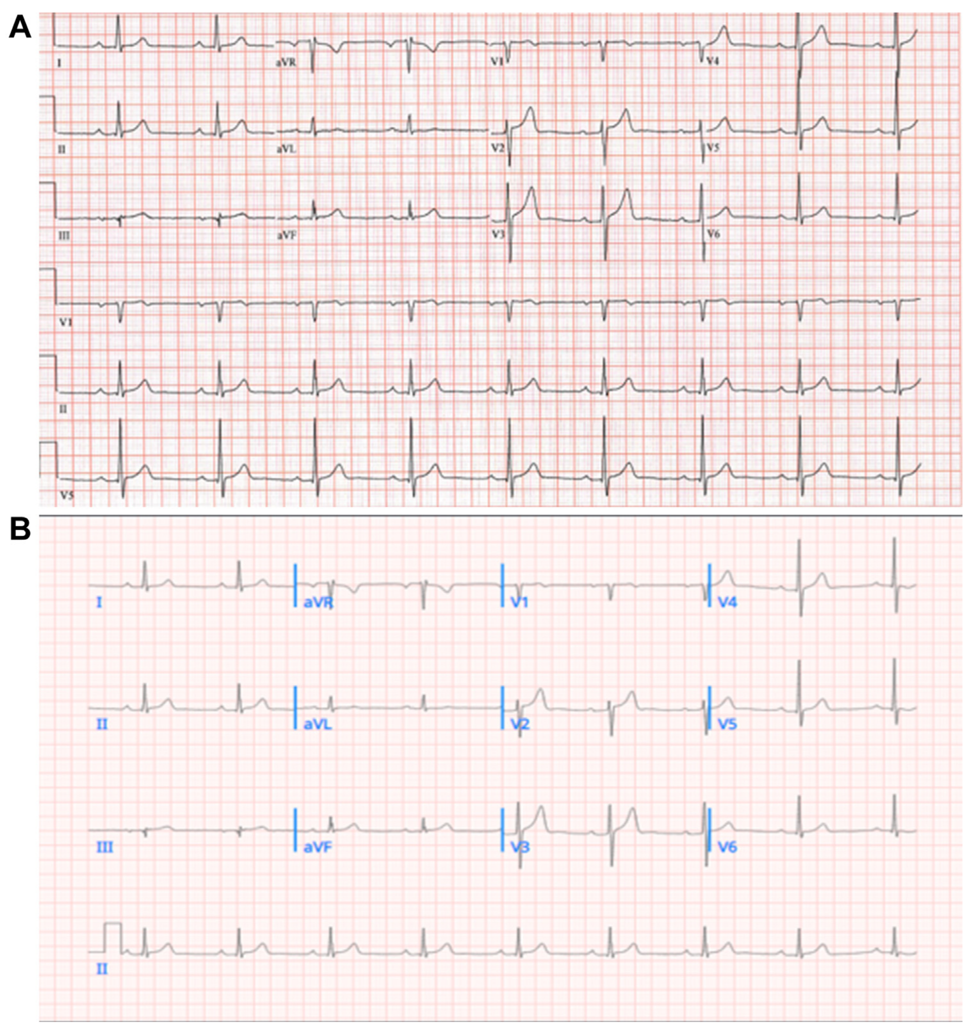
SAM = systolic anterior motion

motion of the mitral valve, which often results in symptomatic heart failure, reduced exercise capacity, and exertional syncope.[1,2] However, there are many challenges in recognition of HCM, with some estimates suggesting that ~85% patients with HCM remain underdiagnosed and misdiagnosed. An earlier and accurate diagnosis of HCM is a major unmet need, and there are major efforts to enhance a clinician's ability to optimally diagnose HCM early. This would potentially enable clinicians to institute optimal therapies and provide appropriate risk stratification to reduce the risk of SCD. There are emerging data on role of artificial intelligence enhanced 12-lead electrocardiogram (AI-ECG) in detection of HCM.[3,4] In the current report, we describe the case of a 61-year-old man who was diagnosed with obstructive HCM using AI-ECG as part of an executive physical with subsequent successful therapeutic response to mavacamten.

CASE HISTORY

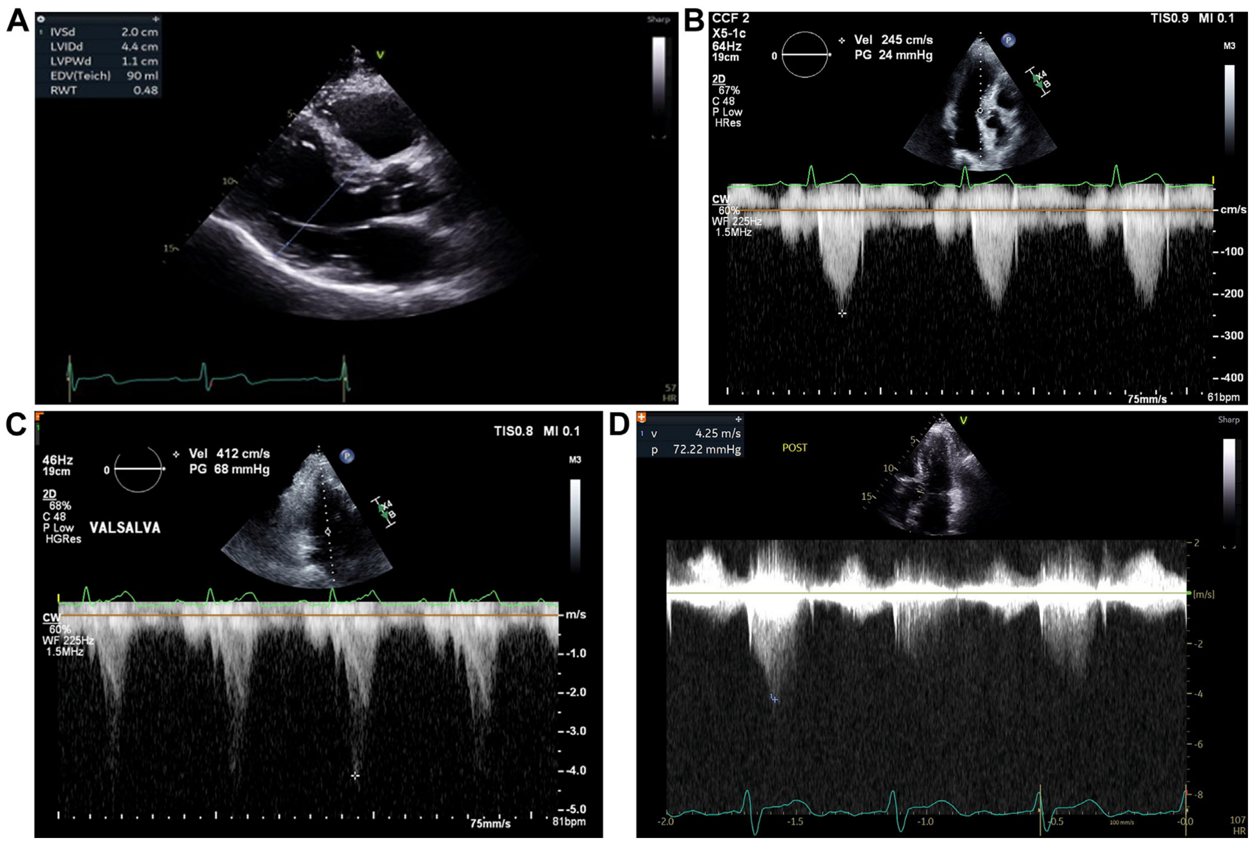
A 61-year-old Caucasian man underwent a physical at the executive health clinic at the Cleveland Clinic. As part of the work-up, he underwent a comprehensive evaluation, which included electrocardiogram (ECG) (Figure 1A), coronary artery calcium (CAC) score, laboratory evaluation, and exercise stress testing. His

FIGURE 1 Baseline Electrocardiographic and Echocardiographic Data



(A) Baseline electrocardiogram (ECG) at initial outpatient visit. (B) Artificial intelligence analyzed ECG.

FIGURE 2 **Baseline Echocardiographic Data**



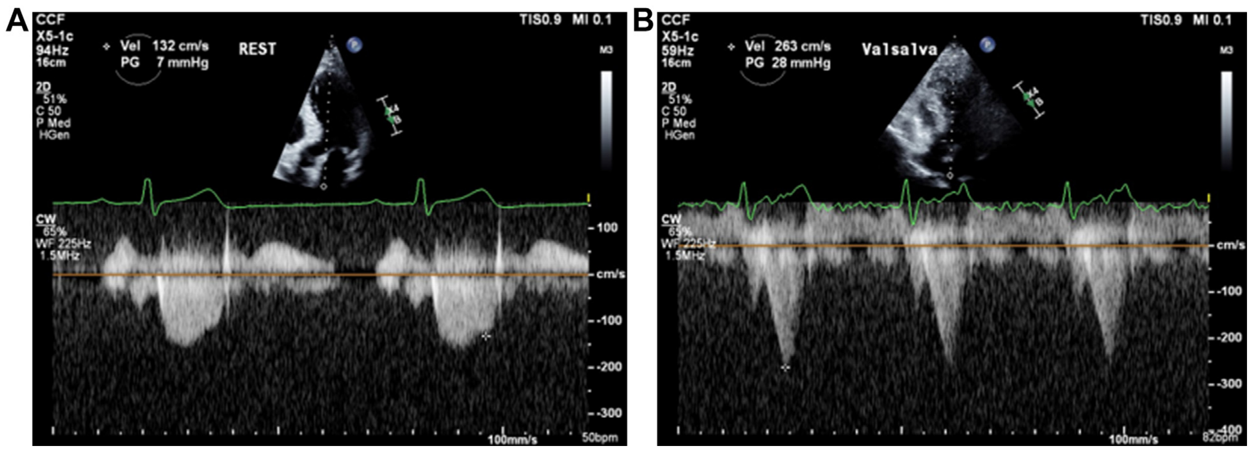
(A) Parasternal long axis view demonstrating severe basal septal hypertrophy with maximal measurement of 2 cm. (B) Spectral Doppler demonstrating no significant left ventricular outflow tract (LVOT) obstruction of 24 mm Hg at rest. (C) Spectral Doppler demonstrating severe LVOT obstruction of 68 mm Hg following Valsalva maneuver. (D) Spectral Doppler demonstrating severe LVOT obstruction of 72 mm Hg postexercise.

baseline ECG was interpreted clinically as normal sinus rhythm with first-degree atrioventricular block; CAC score was 235 (79th percentile for age, gender, and ethnicity), and he achieved 9 metabolic equivalents (METs) on exercise stress test without significant abnormalities. As part of an organization-wide initiative of analyzing all ECGs for HCM, the baseline ECG was subsequently uploaded to a cloud-based platform that uses an AI-ECG algorithm (a convolutional neural network trained algorithm [VIZ-HCM]) to help detect HCM. Gradient weighted class activation mapping (Grad-CAM) was used to create a visualization of key regions (saliency map) in the ECG that contribute to the convolutional neural network-based prediction of HCM. The AI-analyzed ECG (Figure 1B) subsequently received a high probability score (>0.95) for HCM. The patient was subsequently referred to the HCM center for further evaluation.

At the time of initial cardiovascular medicine evaluation, the patient described symptoms of occasional lightheadedness and dizziness with quick positional changes that he had noticed over many years. He also complained of functional incapacity with moderate exertion and excess fatigue but denied chest pain, edema, near syncope, or syncope. There was no history of SCD.

His past medical history was positive for coronary atherosclerosis, mild carotid artery disease, hypertension with hyperaldosteronism, hyperlipidemia, and obstructive sleep apnea. He had no significant surgical history. His medical regimen included metoprolol succinate, eplerenone, losartan, and rosuvastatin. Family history was unremarkable for HCM or sudden cardiac death. Both parents are alive, with a history of hypertension and hyperlipidemia, and 3 siblings are alive and well. He has 2 children

FIGURE 3 **Echocardiographic Data After 4 Weeks of Mavacamten Treatment**



(A) Spectral Doppler demonstrating no significant left ventricular outflow tract (LVOT) obstruction of 7 mm Hg at rest. (B) Spectral Doppler demonstrating no significant LVOT obstruction of 28 mm Hg following Valsalva maneuver.

who are alive and well. He works as an executive and drinks 2 to 3 glasses of alcohol socially weekly and denies illicit drug use. He is physically active and does aerobic exercise 4 to 5 times a week.

Vital signs were normal, as follows: blood pressure of 125/78 mm Hg, heart rate 56 beats/min, resting pulse oximetry of 97%, and body mass index of 28 kg/m^2. Physical examination results were unremarkable, and there was no appreciable murmur at rest or with Valsalva maneuver.

ADDITIONAL BASELINE INVESTIGATIONS

Baseline laboratory values revealed normal chemistry panel, blood count and lipid profile. N-terminal brain natriuretic peptide was normal. The baseline resting echocardiogram revealed severe (2 cm) basal interventricular septal hypertrophy (**Figure 2A**), preserved left ventricular ejection fraction (LVEF) of 60%, moderate systolic anterior motion (SAM) of the mitral valve (Videos 1A and 1B), and resting and post-Valsalva maneuver LVOT gradients of 29 mm Hg and 68 mm Hg, respectively (**Figures 2B and 2C**). Subsequently, the patient underwent treadmill stress echocardiogram achieving 8.8 METs without evidence of ischemia. At peak exercise, there was evidence of severe SAM with septal contact and an LVOT gradient of 72 mm Hg (Video 1C, **Figure 2D**). There was mild late gadolinium enhancement at right ventricular insertion point on contrast-enhanced cardiac magnetic resonance. A 48-hour Holter monitor did not reveal any arrhythmias.

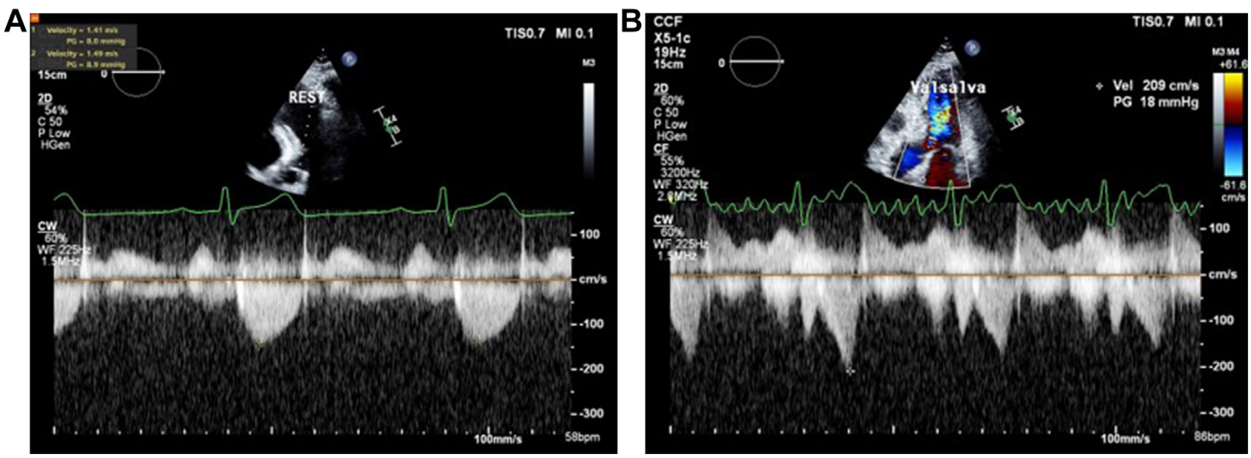
INITIAL MANAGEMENT

The patient was diagnosed with obstructive HCM with severe dynamic LVOT obstruction and NYHA functional class II symptoms, in addition to positional dizziness. Given the history of hyperaldosteronism and well-controlled hypertension, a decision was made to continue current medical therapy, which included eplerenone and losartan in addition to metoprolol. The patient did not yet meet criteria for septal reduction therapy (SRT) and was unwilling to consider surgery as the next option. Subsequently, after appropriate shared decision making and enrollment in the risk-evaluation and mitigation strategy program, he was initiated on mavacamten 5 mg/day. Genetic counseling, testing, and screening of first-degree relatives were recommended. Based on current guidelines, the patient did not meet criteria for a primary prevention internal cardioverter defibrillator.

SUBSEQUENT FOLLOW-UP

As part of the Risk Evaluation and Mitigation Strategy (REMS) program, the patient returned for follow-up limited echocardiograms at Week 4 and 8. At week 4, the LVEF was 61% with mild resting SAM and a resting LVOT gradient of 7 mm Hg (Video 2A, **Figure 3A**). Post-Valsalva maneuver revealed moderate SAM and an LVOT gradient of 28 mm Hg (Video 2B, **Figure 3B**). Similarly, at Week 8, the LVEF was 62% with mild resting SAM and a resting LVOT

FIGURE 4 Echocardiographic Data After 8 Weeks of Mavacamten Treatment



(A) Spectral Doppler demonstrating no significant left ventricular outflow tract (LVOT) obstruction of 9 mm Hg at rest. (B) Spectral Doppler demonstrating severe LVOT obstruction of 18 mm Hg following Valsalva maneuver.

gradient of 9 mm Hg (Video 3A, **Figure 4A**). Post-Valsalva maneuver revealed moderate SAM and an LVOT gradient of 18 mm Hg (Video 3B, **Figure 4B**). At Week 8, his fatigue was slightly improved from baseline. He reports less dizziness with bending over and he reports that he gets less short of breath climbing stairs. He denies chest pain, edema, syncope, nocturnal dyspnea, or orthopnea. He has not had any interval hospitalizations. The patient also underwent genetic counseling and has agreed to undergo genetic testing.

DISCUSSION

In the current case, we highlight the effective clinical implementation of 2 novel aspects of managing this patient with symptomatic obstructive HCM. The first is the use of AI-ECG in making an appropriate diagnosis of obstructive HCM in a patient who was unaware of the actual diagnosis and may not have realized that his various symptoms were likely a result of obstructive HCM. Indeed, there are emerging data regarding the use of AI-ECG in diagnosis of HCM with a high degree of accuracy.[3,4] The second was successful initiation of clinically approved mavacanten, a novel cardiac myosin inhibitor therapy, which has been demonstrated in clinical trials to result in significant longer-term improvement in LVOT obstruction, symptoms, quality of life and reduction in need for SRT.[5,6] Indeed, in the current guidelines, mavacamten is considered a Class I and Class IIa indication in management of symptomatic obstructive HCM in patients who remain symptomatic despite background HCM therapy.[1,2] And we have recently demonstrated favorable clinical and logistic real-world experience of managing patients with symptomatic obstructive HCM using mavacamten.[7]

With concerns regarding gross underdiagnosis and misdiagnosis of HCM, increasing use of AI-ECG has great potential in identifying patients who would have previously remained unaware of their medical conditions, potentially putting them at increased risks associated with progression of disease (eg, heart failure, atrial fibrillation, ventricular arrhythmias, and SCD).[1,2] Furthermore, an earlier and an accurate HCM diagnosis might enable early initiation of highly effective medical therapies such as cardiac myosin inhibitors, which have demonstrated favorable cardiac remodeling in clinical trials.[3,4] This is in addition to providing effective risk stratification to prevent sudden cardiac death and appropriate referral for SRT in selected cases.[1,2] However, for most consumers of new medical information, different AI techniques are too complicated to understand, with a major issue being the "black-box" nature of proprietary software. As a result, as AI techniques are rapidly evolving, we need to ensure that these techniques are based on accurate, high-quality, ethnically diverse, and large-scale training and validation datasets so that we can enhance patient trust and increase adoption by health care professionals.[8]

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KEY WORDS cardiomyopathy, echocardiography, electrocardiogram

APPENDIX For supplemental videos, please see the online version of this paper.